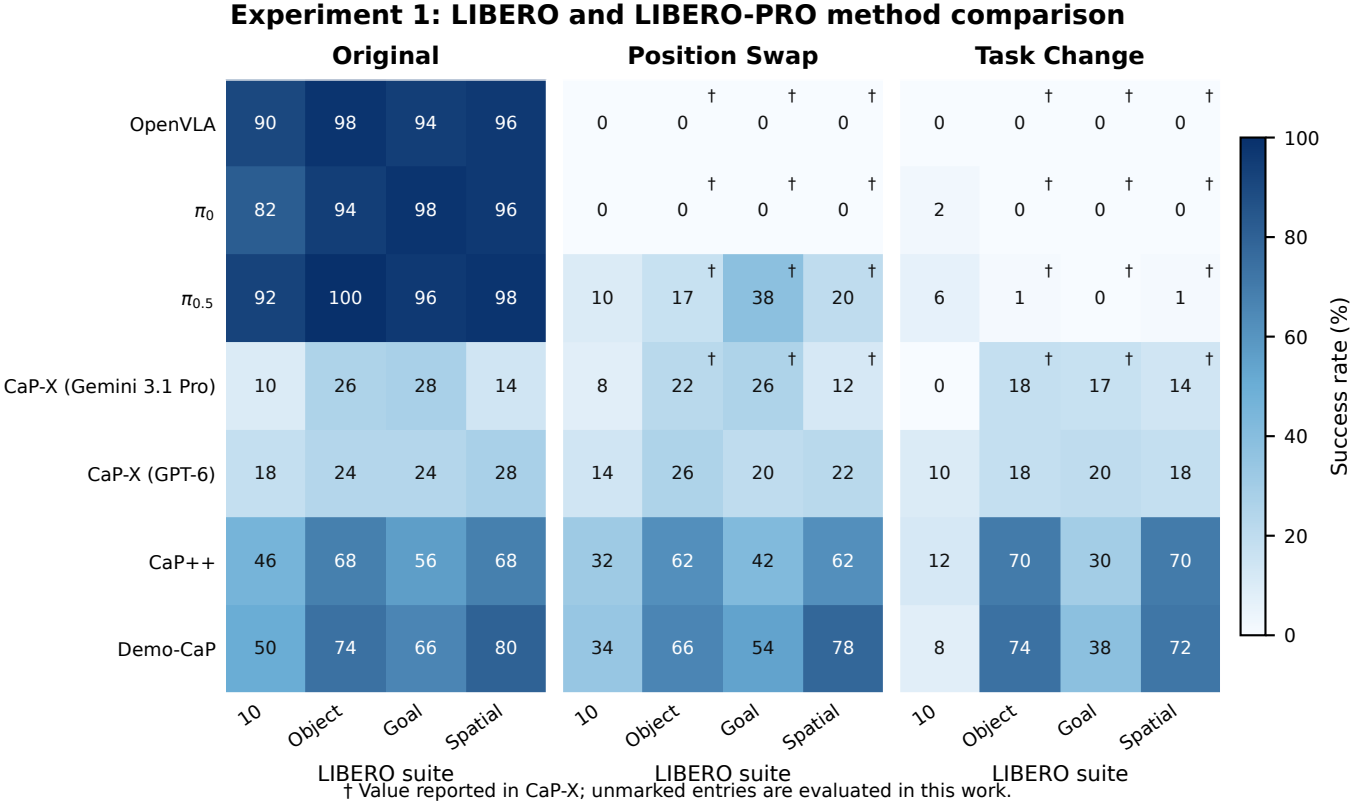

Experiment 1: LIBERO and LIBERO-PRO method comparison
Original
Position Swap
Task Change
OpenVLA
π_0
$\pi_{0.5}$
CaP-X (Gemini 3.1 Pro)
CaP-X (GPT-6)
CaP++
Demo-CaP
90 98 94 96
82 94 98 96
92 100 96 98
10 26 28 14
18 24 24 28
46 68 56 68
50 74 66 80
0 0† 0† 0†
0 0† 0† 0†
10 17† 38† 20†
8 22† 26† 12†
14 26 20 22
32 62 42 62
34 66 54 78
0 0† 0† 0†
2 0† 0† 0†
6 1† 0† 1†
0 18† 17† 14†
10 18 20 18
12 70 30 70
8 74 38 72
10
Object
Goal
Spatial
LIBERO suite
Success rate (%)
100
80
60
40
20
0
† Value reported in CaP-X; unmarked entries are evaluated in this work.